Sushant Lokhande

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Professional Summary

AI Engineer and Published ML researcher with hands-on experience building and evaluating **LLM applications, RAG systems, tool-using agents, ML pipelines, and rapid AI prototypes** using Python, PyTorch, SQL, REST APIs, and structured data. Experienced designing **evaluation datasets, controlled experiments, baselines, failure analysis, retrieval pipelines, and automated validation**.

Technical Skills

Programming & Data: Python, SQL, TypeScript, JavaScript, C++, Pandas, NumPy, JSON, XML
Applied AI: LLMs, RAG, Tool Calling, Multi-Agent Systems, MCP, Structured Outputs, Prompt Design, HITL
ML & Evaluation: PyTorch, Scikit-learn, CNNs, Vision Transformers, Embeddings, Precision, Recall, F1, Baselines
Experimentation: Golden Datasets, Ground Truth, Model Evaluation, Controlled Experiments, Benchmarking, Error Analysis
Engineering: FastAPI, REST APIs, PostgreSQL, Redis, Docker, pytest, CI/CD, Grafana, Prometheus

Experience

San Jose State University | AI Engineer, Applied AI Research — Aug 2025 to May 2026

- **Automated evaluation across 500+ assignments** by architecting a Python multi-agent system that combined retrieval, theory assessment, code execution, defect detection, structured outputs, and automated feedback.
- **Supported 250+ users** by translating grading and validation requirements into testable AI workflows using RAG, reference solutions, rubrics, tool orchestration, and human-review checkpoints.
- **Orchestrated 4 specialized AI stages** for retrieval, reasoning, code validation, and reporting by designing parallel agent workflows with explicit failure handling and reproducible intermediate state.
- **Benchmarked 500+ AI-assisted evaluations** by comparing generated outputs against reference criteria, investigating failure modes, and converting recurring errors into retrieval, prompt, and validation improvements.

Turito | Software Engineer Intern, AI & Data Platform — Jun 2022 to Oct 2022

- **Supported 70K+ users** by developing Python APIs, data-processing workflows, validation logic, and customer-facing application services across learning and operational workflows.
- **Reduced API latency from 8s to 1.2s** by profiling request paths, optimizing PostgreSQL queries, introducing Redis caching, and eliminating redundant processing.
- **Processed 2M+ application events per day** by developing asynchronous ingestion, transformation, persistence, and analytics workflows across production backend services.
- **Reduced production regressions by 38%** by expanding pytest coverage, validating API and database behavior, containerizing environments, and turning recurring defects into automated regression tests.

Trivia Softwares | Machine Learning Intern — Nov 2021 to Jan 2022

- **Processed 500K+ records** by building Python/Flask ML and data-processing services with Pandas, NumPy, SQL, reusable preprocessing components, and REST APIs.
- **Validated ML workflows across 500K+ records** by implementing input checks, deterministic preprocessing, structured error handling, regression tests, and reproducible Docker environments.

Projects

Reel Rank – AI Content Discovery Platform | *Python, React, TypeScript, FastAPI, PyTorch, Embeddings*

- **Improved withheld-item recall by approximately 10x** over a collaborative-only baseline by combining semantic retrieval, two-tower recommendations, ANN search, metadata features, and neural reranking.
- **Combined 3 ranking stages** into an end-to-end content discovery workflow by integrating natural-language queries, semantic retrieval, personalized ranking, and continuously refreshed movie metadata.
- **Automated a 24-hour catalog refresh cycle** by building ingestion and synchronization workflows for external content metadata while preserving cold-start and temporal evaluation behavior.

Research and Publications

Image-Based Techniques and Ensemble Soft Voting for Malware Classification — 2026

- **Evaluated 8 input representations across 50K+ samples** by benchmarking HOG, Haralick features, CNN representations, VGG16, ResNet50, and Vision Transformer embeddings across 17 malware families.
- **Achieved 80.2% test accuracy** by designing a 15-model soft-voting ensemble that improved classification performance by **2.4 percentage points** over the strongest individual model.
- **Benchmarked 15 model variants against consistent test data** by standardizing preprocessing, evaluation criteria, controlled comparisons, and failure analysis across classical and deep-learning approaches.

Education

San Jose State University | M.S. in Computer Science — May 2026